



Amazon Product Descriptions Word Cloud

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 Stakeholders	

This task involves you developing a web crawler that generates a word cloud of the most interesting words in the product descriptions people view on Amazon.

The system will get requests with Amazon URLs of products (for example: <http://www.amazon.com/gp/product/B00VVOCSOU>) through a REST endpoint that you need to develop, will go to these pages and extract the product descriptions for it and create a word cloud with the most common and interesting words in the descriptions.

Size:48-Inch

Product Description

Enjoy The Creative Life with the TCL 48" 1080p direct LED HDTV. It delivers premium picture quality and tremendous value in a sophisticated slim frame design perfect for bringing entertainment to any space. This flat screen LED HDTV features High Definition 1080p resolution for a sharper image and TCL True Color Technology for brilliant color and contrast. With direct LED backlighting, view darker blacks and luminous brightness while maintaining the best standards in energy efficiency.

Example of a product description. Please notice that the description is the section that starts with *"Enjoy The Creative Life"*



Technical Details

1. You can assume that the load of the of the requests is about 1/sec, but make sure that your design can support a load several orders of magnitudes larger than that.
2. The list of URLs will be provided through a set of CURL calls. Please use **simulateRequests.sh** script to simulate those requests while you develop. For example you can run it as follows: `./simulateRequests.sh localhost 8080`

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3. URLs might repeat themselves, and as with any good engineering system, we would like to avoid fetching the same page over and over again.
4. Assume the word corpus can be very large, so a naive calculation of all the top words in the corpus will not be quick enough to render promptly on the page. You will need to address it either through the algorithm, data structures or technology.
5. We would want to avoid having very common words in the word cloud (e.g. a, the, is).
6. This is not a frontend development assignment, please don't spend too much time on making the word cloud pretty. It's ok, if it's even a textual representation of the data powering the word cloud.

Assignment Review Details

This assignment is a test of your systems engineering skills more than your coding skills. Therefore while we would like to see clean and organized code, it is more important that the system is built with the right components, can scale well, and you will be able to explain every design decision that you have made building this system.